

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2021

Bachelor in Computer Applications

Course Title: **Software Engineering**

Code No: **CACS253** Semester: **IV**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. What is software engineering? Explain the challenges facing software engineering.
3. What is extreme programming? Explain how the principles underlying extreme programming lead to the accelerated development of software.
4. Define quality assurance? Under what circumstances would you recommend the use of staged representation of the CMM?
5. What is architectural design? Explain layered model of software architecture with example.
6. Differentiate between verification and validation. Explain why software inspection is an effective technique for discovering errors in software?
7. What is software reengineering? List its advantages. Explain source code translation and reverse engineering approach in brief.
8. What do you mean by software quality management? Explain software quality management activities in brief.

Group C

Attempt any TWO questions[2x10=20]

9. What is software process model? List different software process model. Explain how both the waterfall model of the software process and prototyping model can be accommodated in the spiral process model.
10. What do you mean by software requirement? Differentiate between functional requirement and non-functional requirement. List the functional requirements of online examination system and represent it in use case diagram.
11. a) What is risk management? Explain risk management process in software engineering with block diagram. b) List good programming practices. Explain CASE and its type with example.